

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 3 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|------------------------------|---|
| 1. Subspace _____ | (A) The number of vectors in a basis (e.g. r for $C(A)$, $n - r$ for $N(A)$). |
| 2. Nullspace $N(A)$ _____ | (B) The span of the columns of A ; the reachable outputs $\mathbf{b}$, a subspace of $\mathbb{R}^m$. |
| 3. Column space $C(A)$ _____ | (C) A set in which only the all-zero combination gives $\mathbf{0}$ (no vector is a combination of the others). |
| 4. Special solution _____ | (D) The set of all solutions to $A\mathbf{x} = \mathbf{0}$; a subspace of $\mathbb{R}^n$. |
| 5. Linear independence _____ | (E) The number of pivots r ; equals both the row rank and the column rank. |
| 6. Basis _____ | (F) A subset closed under addition and scalar multiplication (so it contains $\mathbf{0}$). |
| 7. Dimension _____ | (G) An independent set that spans the space — the fewest vectors that build every vector in it. |
| 8. Rank _____ | (H) A solution of $A\mathbf{x} = \mathbf{0}$ found by setting one free variable to 1 and the rest to 0. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- Which set is **not** a subspace of $\mathbb{R}^2$? (A) the line $y = 2x$ (B) all of $\mathbb{R}^2$ (C) the line $y = 2x + 1$ (D) $\{\mathbf{0}\}$
- For an $m \times n$ matrix of rank r , the nullspace $N(A)$ has dimension (A) r (B) $n - r$ (C) $m - r$ (D) n
- The columns of A are linearly independent exactly when (A) $N(A) = \{\mathbf{0}\}$ (B) $C(A) = \mathbb{R}^m$ (C) A is square (D) A has a zero row
- For *any* matrix, the row rank and the column rank are (A) always equal (B) equal only if A is square (C) never equal (D) unrelated
- When $A\mathbf{x} = \mathbf{b}$ is solvable, its complete solution has the form (A) $\mathbf{x}_p$ only (B) $\mathbf{x}_n$ only (C) $\mathbf{x}_p + \mathbf{x}_n$ (one particular + the whole nullspace) (D) always a single point
- If $\mathbf{x}$ is in the nullspace of A , then $\mathbf{x}$ is perpendicular to (A) every column of A (B) every row of A (C) the vector $\mathbf{b}$ (D) nothing

Part C — Short Answer & Computation (35 pts)

Show your work. Problems 2–7 all use the same matrix $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 2 & 5 \end{bmatrix}$.

1. Use the **closure test** to decide whether $W = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$ is a subspace of $\mathbb{R}^3$. Check all three conditions (contains $\mathbf{0}$; closed under addition; closed under scalar multiplication).
2. Row-reduce A and find every **special solution** of $A\mathbf{x} = \mathbf{0}$. State $\dim N(A)$.
3. Solve $A\mathbf{x} = \mathbf{b}$ **completely** for $\mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$. Write your answer as $\mathbf{x} = \mathbf{x}_p + (\text{nullspace part})$.
4. Is $\mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ in the column space $C(A)$? Show how you know (reduce $[A \mid \mathbf{b}]$ and read off solvability).
5. Are the columns of A linearly independent? Give a **basis** for $C(A)$ and state its dimension (the rank r).
6. A is 3×3 with rank $r = 2$. State the dimensions of all **four** subspaces: $C(A)$, $N(A)$, the row space $C(A^T)$, and the left nullspace $N(A^T)$. Which two live in the input space $\mathbb{R}^3$ and which two in the output space $\mathbb{R}^3$?
7. Verify that the special solution $\mathbf{s} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$ is perpendicular to *every* row of A (compute the three dot products). Which theorem does this illustrate?

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

1. Consider the line $W = \{(x, y) \in \mathbb{R}^2 : y = 2x + 1\}$. **(a)** Does W contain the zero vector? **(b)** Is W a subspace of $\mathbb{R}^2$? Justify with the closure test. **(c)** Contrast W with the line $y = 2x$: which is a subspace, and why?
2. For the matrix A above (rank 2, special solution $\mathbf{s} = (-1, -1, 1)$): **(a)** Explain why $N(A)$ and the row space are perpendicular, using the idea that “ $A\mathbf{x} = \mathbf{0}$ says each row $\cdot \mathbf{x} = 0$.” **(b)** The two subspaces have dimensions 2 and 1 inside $\mathbb{R}^3$. Explain why together they “fill” $\mathbb{R}^3$ and meet only at $\mathbf{0}$.